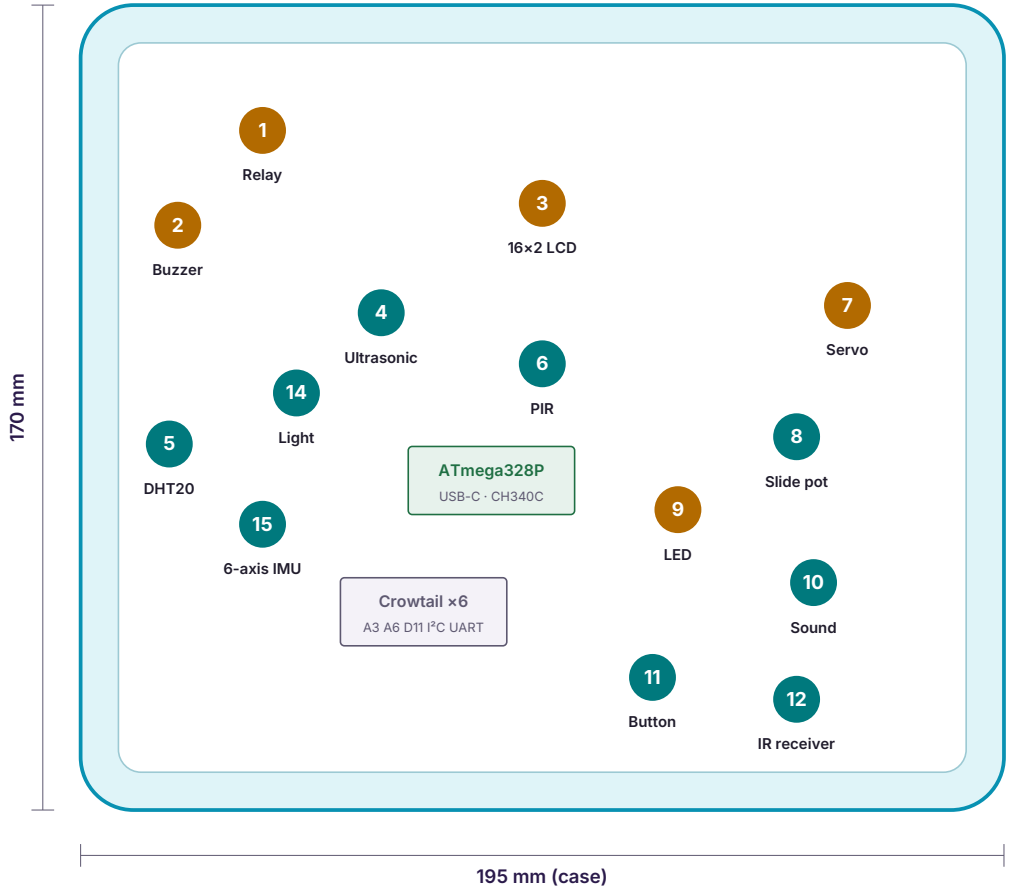
01 STARTER BOARD AND CASE

STARTER BOARD IN ITS CASE · TOP VIEW · POSITIONS INDICATIVE



- Sensor (reads)
- Actuator (moves)
- Controller
- Expansion ports

Height with lid closed: 46 mm

Weight: 380 g

Case material: PP

USB-C on the board edge

#	Module	Pin
1	Relay	D4
2	Buzzer	D3
3	16×2 LCD	I²C 0x21
4	Ultrasonic	D6 / D5
5	DHT20	I²C 0x38
6	PIR	A2
7	Servo	D9
8	Slide pot	A0
9	LED	D10
10	Sound	A1
11	Button	D7
12	IR receiver	D2
14	Light	I²C 0x5C
15	6-axis IMU	I²C 0x6B/0x68

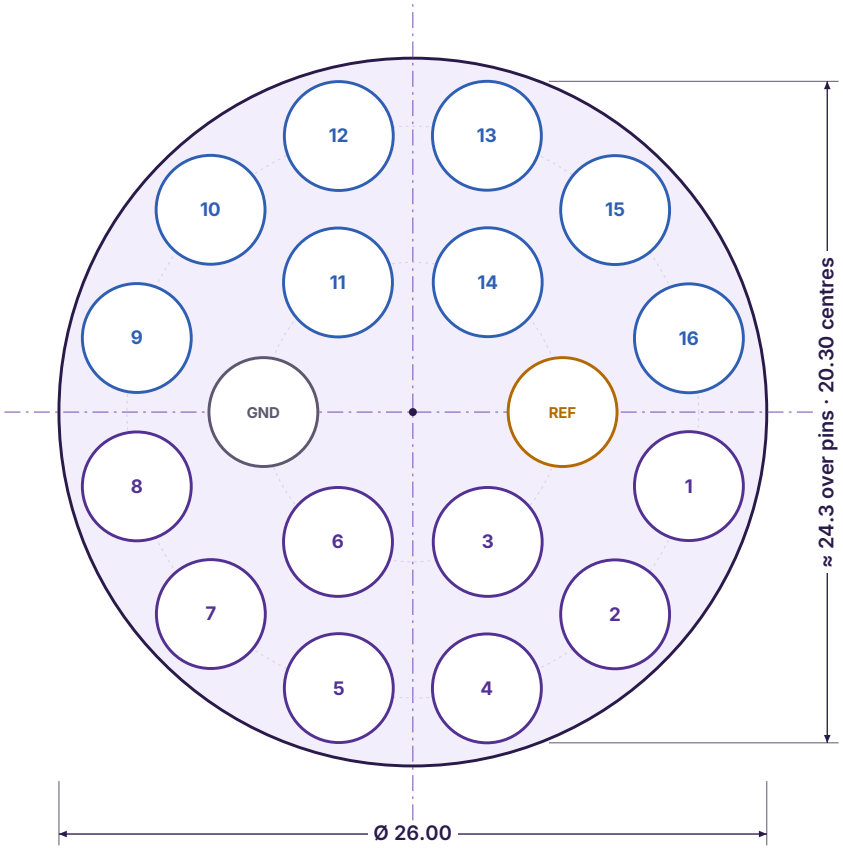
Title	OctoBio Kit — starter board layout
Doc. No.	OB-MD-002
Revision	Rev. A · 10 Nov 2026
Board	Elecrow SEA0004T V1.0
Units	mm

Figure 1 — Module positions are indicative, taken from product photos; the outline is the carrying case (195 × 170 × 46 mm). Pins: OB-PIN-004.

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02 OCTOPUS 16 OUTLINE

PIN-SIDE VIEW · +y toward the stacking connector

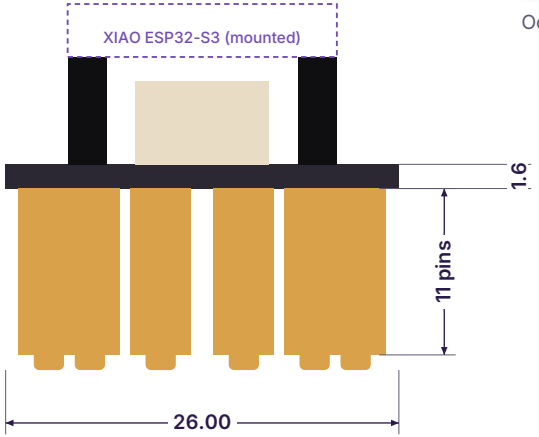


Pitch ≈ 5.38 (hexagonal)

Inner ring r ≈ 5.5: REF, GND, ch 3, 6, 11, 14

Outer ring r ≈ 10.5: remaining 12 channels

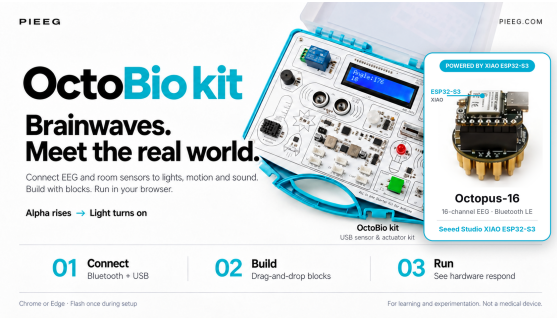
SIDE ELEVATION (schematic)



Pogo travel compresses under strap pressure; socket and XIAO heights depend on the parts fitted and are shown for orientation only.

Notes

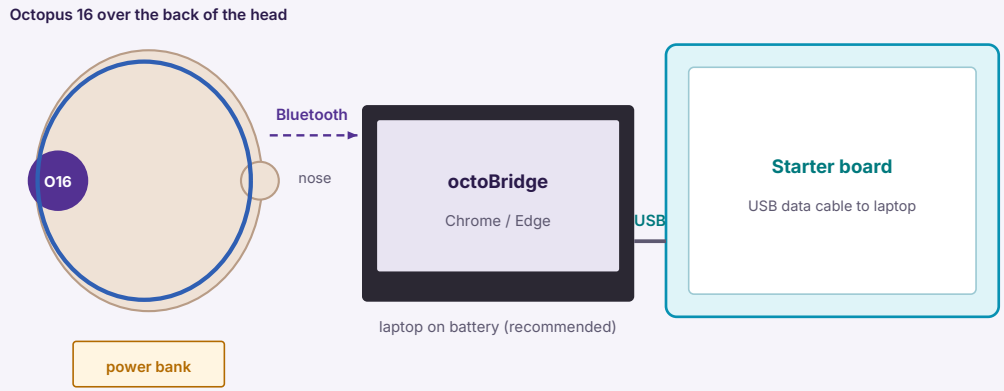
- Round board Ø 26.00, 1.6 mm, 4 layers.
- 18 pogo pins Ø4 × 11 mm on a hexagonal ≈ 5.38 mm grid.
- REF at +5.50, 0 and GND at −5.49, 0 (inner ring).
- XIAO ESP32-S3 plugs into two 7-pin sockets on the component side — match the 5V / GND / 3V3 labels.
- No mounting holes: held by a strap or headband with the power bank.



Octopus 16 with its XIAO ESP32-S3.

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03 CLASSROOM SETUP



Only the starter board is cabled. The Octopus 16 on the head stays wireless and runs from its own power bank.

Figure 3 — One desk: the student wears the Octopus 16; only the starter board is cabled to the laptop.



Octopus 16 over the back of the head, power bank under the strap.

Item	Placement	Why
Octopus 16	Back of the head (visual cortex), pressed straight down	Alpha shows up clearly; every pin must touch the scalp
Power bank	Under the strap or in a pocket	Battery only while worn — no cable to the computer
Laptop	Within a few metres, on battery if possible	Bluetooth range; less 50/60 Hz hum
Starter board	On the desk, lid open, USB data cable to the laptop	Leave room for the servo arm and keep liquids away

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